A recipe by itself cannot make soup. You need a person to read the recipe and follow the prescribed set of steps. But you can build a machine that embodies this algorithm and follows it automatically. Then you just need to provide the machine with water, electricity and vegetables – and it will prepare the soup by itself. There aren't many soup machines around, but you are probably familiar with beverage vending machines. Such machines usually have a slot for coins, an opening for cups, and rows of buttons. The first row has buttons for coffee, tea and cocoa. The second row is marked: no sugar, one spoon of sugar, two spoons of sugar. The third row indicates milk, soya milk, no milk. A man approaches the machine, inserts a coin into the slot and presses the buttons marked 'tea', 'one sugar' and 'milk'. The machine kicks into action, following a precise set of steps. It drops a tea bag into a cup, pours boiling water, adds a spoonful of sugar and milk, and ding! A nice cup of tea emerges. This is an algorithm.[17]

Over the last few decades biologists have reached the firm conclusion that the man pressing the buttons and drinking the tea is also an algorithm. A much more complicated algorithm than the vending machine, no doubt, but still an algorithm. Humans are algorithms that produce not cups of tea, but copies of themselves (like a vending machine which, if you press the right combination of buttons, produces another vending machine).

The algorithms controlling vending machines work through mechanical gears and electric circuits. The algorithms controlling humans work through sensations, emotions and thoughts. And exactly the same kind of algorithms control pigs, baboons, otters and chickens. Consider, for example, the following survival problem: a baboon spots some bananas hanging on a tree, but also notices a lion lurking nearby. Should the baboon risk his life for those bananas?

This boils down to a mathematical problem of calculating probabilities: the probability that the baboon will die of hunger if he does not eat the bananas, versus the probability that the lion will catch the baboon. In order to solve this problem the baboon needs to take into account a lot of data. How far am I from the bananas? How far away is the lion? How fast can I run? How fast